

Intelligent Coarse Dropout and Anti-ICD: Saliency-Guided Masking Augmentation for Visual Classifiers

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Abstract

Visual classifiers trained with standard augmentation pipelines tend to exploit spurious correlations: background statistics, texture patterns, or scene-level cues that happen to co-occur with class labels but do not reflect object structure. Conventional augmentation strategies, including random crops, colour jitter, and policy-based methods such as RandAugment and AutoAugment, are constructed without reference to what the model has actually learned, and so offer no mechanism for targeting the specific shortcuts a given classifier has acquired.

We present Intelligent Coarse Dropout (**ICD**) and its complement Anti-ICD (**AICD**), two data augmentation methods that condition spatial masking on the model’s own class activation maps. Standard coarse dropout (cutout) hides rectangular regions chosen at random, erasing background as readily as object. ICD and AICD make this saliency-aware: they split the image into a grid of tiles, score each tile by its mean saliency, and selectively mask tiles by a percentile threshold. ICD masks the highest-saliency tiles, the regions the model already relies on, so that the classifier must recruit additional cues. AICD masks the lowest-saliency tiles, perturbing the background while leaving the discriminative region intact. Masked tiles are not filled with a hard black box but with one of several soft strategies (blurred original, local mean, noise, or a constant), which preserve different aspects of local context.

We describe the formal construction of both masks, the family of fill strategies, the role of the tile size and threshold hyperparameters, and a reference implementation in the open-source BNNR library. This paper introduces the methods; a quantitative evaluation against standard augmentation baselines is left to a companion study.

1 Introduction

Convolutional neural networks trained on natural image datasets develop systematic biases toward low-level texture features over global shape [Geirhos et al., 2019]. On fine-grained datasets the effect is pronounced: classifiers readily latch onto contextual cues such as the grass behind a terrier, the greenhouse behind a flower, or the sky behind an aircraft. These correlations hold in the training distribution and can yield high validation accuracy, yet they transfer poorly to images where the shortcut is absent and leave the model brittle under distribution shift.

Data augmentation is the standard remedy. The prevailing approach applies a fixed family of photometric and geometric transformations uniformly throughout training. AutoAugment [Cubuk et al., 2019] and its descendants, RandAugment [Cubuk et al., 2020] and TrivialAugment [Müller and Hutter, 2021], search over large transformation spaces but treat the network as a black box: the policy is fixed before training and never consults the model’s internal state. Spatial-masking regularisers such as Cutout [DeVries and Taylor, 2017] and Random Erasing [Zhong et al., 2020] remove rectangular regions to discourage reliance on any single patch, but the location of the mask is chosen uniformly at random. A random mask is blind to content; it deletes empty background as often as it deletes the object.

This raises a natural question. If a saliency method can tell us which regions the model currently treats as discriminative, can we place the mask there deliberately? KeepAugment [Gong et al., 2021] explores one side of this idea by *protecting* high-saliency regions from augmentation, on the premise that preserving informative content helps accuracy. We examine the complementary operations. What happens when the high-saliency regions are *removed*, forcing the classifier to find evidence elsewhere? And what happens when only the low-saliency regions, the background and irrelevant texture, are removed, concentrating the perturbation away from the object?

We formalise both operations as augmentation methods. **Intelligent Coarse Dropout (ICD)** masks the highest-saliency tiles of each input; **Anti-ICD (AICD)** masks the lowest-saliency tiles. Both derive their masks from a GradCAM [Selvaraju et al., 2017] saliency map evaluated at the model’s current parameters, so the augmentation tracks what the model has learned at each stage of training. Crucially, the masking decision is made at the level of coarse tiles rather than individual pixels: the image is divided into an 8×8 -pixel grid (by default), each tile is scored by its mean saliency, and tiles are masked by a percentile threshold. Coarse tiling makes the mask robust to the high-frequency noise that pixel-level saliency maps often exhibit, while keeping the masked region spatially coherent. Rather than a hard black rectangle, masked tiles are filled with a soft strategy: a blurred copy of the original, a local or global mean colour, matched noise, or a constant. The surrounding context then degrades gracefully.

The contributions of this paper are as follows.

1. We introduce ICD and AICD, two saliency-conditioned coarse-dropout augmentations, and give the formal tile-based mask construction together with the percentile-threshold rule that distinguishes them.
2. We describe five fill strategies for masked tiles and characterise how each alters the information that remains available to the model.
3. We discuss the methods’ expected behaviour in terms of curriculum learning and spurious-correlation removal, and situate them with respect to random masking and saliency-protecting augmentation.
4. We document a reference implementation in the open-source BNNR library [Walo et al., 2026], including the use of a precomputed saliency cache that makes online application practical.

This paper is concerned with the methods themselves. A controlled empirical comparison of ICD and AICD against RandAugment, TrivialAugment, and AutoAugment, together with the accompanying benchmark protocol, is the subject of a separate study.

2 Related Work

Masking augmentations. Cutout [DeVries and Taylor, 2017] removes a fixed-size square at a random location and was among the first to show that occluding contiguous regions acts as an effective regulariser. Random Erasing [Zhong et al., 2020] generalises this to rectangles of random aspect ratio and area, optionally filled with noise or the dataset mean. Both choose the masked region without regard to image content. ICD can be viewed as a content-aware Cutout: the shape of the occlusion follows the model’s saliency rather than a random box, and the special case of solid zero-fill on the high-saliency tiles recovers a saliency-targeted form of classical Cutout. CutMix [Yun et al., 2019] replaces a region with a patch from a second image and mixes the labels accordingly; unlike CutMix, ICD and AICD are single-image operations and leave the label unchanged.

Saliency-aware augmentation. The closest line of work conditions augmentation on a saliency or attention signal. KeepAugment [Gong et al., 2021] computes a saliency map and *preserves* the most important region, restricting augmentation to the remainder, with the goal of not destroying discriminative information. ICD adopts the opposite stance: it deliberately occludes the high-saliency region to discourage over-reliance on it, while AICD occupies the same regime as KeepAugment by perturbing only the low-saliency background. The two methods thus bracket the design space that KeepAugment occupies on one side. Other saliency-driven approaches operate through mixing rather than dropout: SaliencyMix [Uddin et al., 2021] uses a saliency map to choose the source patch in a CutMix-style operation, and SnapMix [Huang et al., 2021] uses class activation maps to assign mixed-label proportions in fine-grained recognition. These methods require a second image and modify the target; ICD and AICD do neither.

Class activation maps. ICD and AICD rely on a saliency map to score image regions. We use GradCAM [Selvaraju et al., 2017], which weights the activations of a target convolutional layer by the gradient of the predicted class score and produces a coarse localisation map. Numerous refinements exist, including GradCAM++ [Chattopadhyay et al., 2018] for improved localisation of multiple instances and gradient-free variants such as EigenCAM [Muhammad and Yeasin, 2020], and any of these may in principle drive the masking. Our reference implementation builds on the `pytorch-grad-cam` library [Gildenblat and contributors, 2021], which unifies these methods behind a common interface.

Automated augmentation policies. A parallel body of work searches for augmentation policies rather than conditioning on the model state. AutoAugment [Cubuk et al., 2019] learns a policy by reinforcement learning; RandAugment [Cubuk et al., 2020] reduces the search to two scalar hyperparameters; and TrivialAugment [Müller and Hutter, 2021] dispenses with search entirely, sampling a single transformation per image uniformly. These methods are complementary to ICD and AICD: they define *which* photometric or geometric transformation to apply, whereas ICD and AICD define *where* to apply a content-aware occlusion. The two can be composed.

3 Method

ICD and AICD share a common construction. Given a training image and the classifier in its current state, we compute a saliency map, reduce it to a grid of tile scores, threshold those scores to obtain a binary mask, and replace the masked tiles with a fill. The two methods differ only in the direction of the threshold. We describe each step in turn.

3.1 Saliency Map

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^C$ be a classifier with parameters θ over C classes, and let l index a chosen convolutional layer with feature maps $A^{k,l} \in \mathbb{R}^{h_l \times w_l}$, $k = 1, \dots, K_l$. Each image carries a target class y at which the saliency is computed; by default y is the ground-truth label of the training image, though the model’s prediction $\arg \max_c f_\theta(x)_c$ may be substituted. For an input $x \in \mathbb{R}^{H \times W \times 3}$ the GradCAM [Selvaraju et al., 2017] saliency map is

$$S_\theta(x, y) = \text{ReLU} \left(\sum_{k=1}^{K_l} \alpha_k^y A^{k,l} \right), \quad \alpha_k^y = \frac{1}{h_l w_l} \sum_{i=1}^{h_l} \sum_{j=1}^{w_l} \frac{\partial f_\theta(x)_y}{\partial A_{ij}^{k,l}}. \quad (1)$$

The rectifier discards activations that argue against class y . The resulting map is upsampled bilinearly to the input resolution (H, W) and rescaled to $[0, 1]$; we write $S_\theta(x, y) \in [0, 1]^{H \times W}$ for this normalised map. The reference implementation (Section 5) applies the eigen-smoothing variant of this map by default, taking the saliency as the first principal component of the gradient-weighted activations rather than their plain sum, which suppresses high-frequency artefacts. Either form may be used to drive the masking that follows.

3.2 Tile Scoring

Pixel-level saliency maps are noisy, and masking individual pixels would produce a fragmented, high-frequency occlusion. We instead aggregate the map over a regular grid. Fix a tile size $t \in \mathbb{Z}_{>0}$ and form $n_r = \lfloor H/t \rfloor$ rows and $n_c = \lfloor W/t \rfloor$ columns of non-overlapping $t \times t$ tiles, discarding any boundary strip that does not divide evenly. The score of tile (a, b) is the mean saliency over its pixels,

$$T_{ab} = \frac{1}{t^2} \sum_{p=at}^{(a+1)t-1} \sum_{q=bt}^{(b+1)t-1} [S_\theta(x, y)]_{pq}, \quad a \in \{0, \dots, n_r-1\}, \quad b \in \{0, \dots, n_c-1\}, \quad (2)$$

yielding a score matrix $T \in \mathbb{R}^{n_r \times n_c}$.

3.3 Thresholding: ICD and AICD

Let $\rho \in (0, 1)$ be a percentile parameter and let $Q_\rho(T)$ denote the ρ -quantile of the flattened tile scores $\{T_{ab}\}$. The two methods threshold T in opposite directions.

ICD (Intelligent Coarse Dropout) masks the tiles whose saliency exceeds the ρ -quantile, that is, the most salient tiles:

$$M_{ab}^{\text{ICD}} = \mathbf{1}[T_{ab} > Q_\rho(T)]. \quad (3)$$

Table 1: Fill strategies for masked tiles. Each defines the function $\phi(\cdot)$ in Eq. (6).

Strategy	$\phi(x)$ on masked pixels	Effect
gaussian_blur	blurred copy of x , $\sigma \propto \max(H, W)$	removes detail, keeps coarse colour (default)
local_mean	mean colour of neighbouring unmasked tiles	smooth, locally consistent fill
global_mean	per-channel global mean of x	flat region, no local cue
noise	Gaussian noise matched to per-channel stats	high-frequency perturbation
solid	constant value (default 0)	hard masking; recovers Cutout for M^{ICD}

AICD (Anti-ICD) masks the tiles whose saliency falls below the $(1-\rho)$ -quantile, that is, the least salient tiles:

$$M_{ab}^{\text{AICD}} = \mathbf{1}[T_{ab} < Q_{1-\rho}(T)]. \quad (4)$$

Both masks select, in expectation, the same fraction $1 - \rho$ of all tiles: ICD takes the top $1 - \rho$, AICD the bottom $1 - \rho$. With the default $\rho = 0.70$ each method masks roughly 30% of the tiles. This symmetry is deliberate, since it lets the two methods be compared at matched occlusion area. The masks are exact complements only in the limiting case $\rho = 0.5$; for $\rho > 0.5$ a neutral band of mid-saliency tiles is left untouched by both.

The binary tile mask is expanded to pixel resolution by repeating each tile decision across its $t \times t$ block,

$$\widehat{M}_{pq} = M_{\lfloor p/t \rfloor, \lfloor q/t \rfloor}, \quad (5)$$

with any discarded boundary strip left unmasked, giving $\widehat{M} \in \{0, 1\}^{H \times W}$.

3.4 Fill

A masked tile must be replaced by something. Rather than the hard black box of classical Cutout, we substitute a fill $\phi(x) \in \mathbb{R}^{H \times W \times 3}$, applied only inside the mask:

$$\tilde{x} = (1 - \widehat{M}) \odot x + \widehat{M} \odot \phi(x), \quad (6)$$

where $\odot$ is element-wise multiplication broadcast over channels. We provide five choices of ϕ , summarised in Table 1. The default, **gaussian_blur**, replaces masked pixels with a strongly blurred copy of the image (kernel size set to the nearest odd integer to $0.15 \cdot \max(H, W)$); this removes local detail while retaining coarse colour and luminance, so the occlusion is visible to the network without introducing a sharp artificial edge. The **solid** strategy with a zero value recovers classical Cutout [DeVries and Taylor, 2017], here restricted to the saliency-selected tiles.

3.5 Illustration

Figure 1 shows the full construction on four images. The second column is the GradCAM map of Eq. (1); the third applies ICD, blurring the region the model attends to; the fourth applies AICD, blurring everything else. The same saliency map drives both augmentations. In the Labrador and cat rows the network localises the face, so ICD blurs the face and AICD blurs the surrounding fur and background; in the espresso row the saliency concentrates on the cup rim, and in the sports-car row on the lower body of the car, with the two augmentations partitioning the image accordingly.

Figure 2 fixes the mask and varies the fill, illustrating the qualitative difference between the strategies of Table 1 for both ICD (top) and AICD (bottom). Figure 3 varies the tile size t : small tiles trace the saliency contour closely but fragment the occlusion, whereas large tiles produce a coarse, block-structured mask that is less sensitive to saliency noise.

3.6 Hyperparameters

The construction has three principal hyperparameters. The percentile ρ sets the masked fraction $1 - \rho$: larger ρ masks fewer, more selective tiles. The tile size t trades spatial fidelity for robustness, as in Figure 3. The fill ϕ determines what the network sees in place of the occluded content. In addition, each augmentation is applied stochastically with probability p per image, returning the image unchanged with probability $1 - p$; this keeps a fraction of clean images in every batch. The library defaults are $t = 8$, Gaussian-blur fill, and a percentile of $\rho = 0.70$; we set $p = 0.5$ for training so that half of each batch is

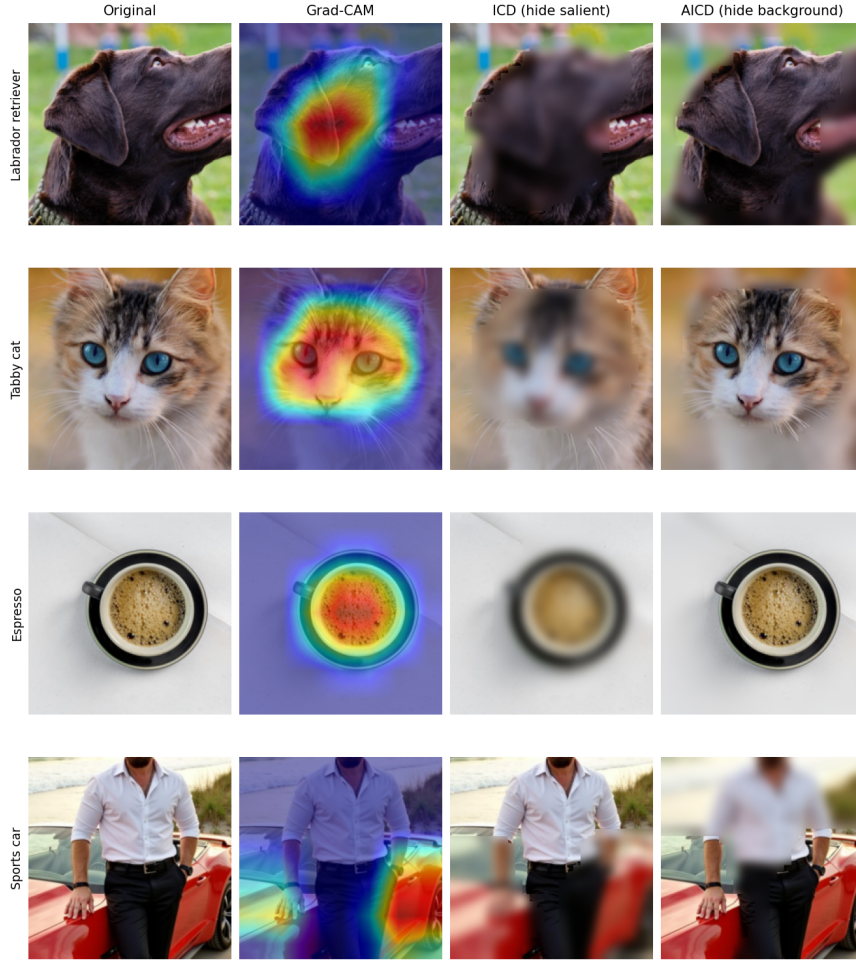


Figure 1: The ICD/AICD construction on four ImageNet classes. From left to right: the original image; the GradCAM saliency map at the final convolutional layer of a ResNet-18; ICD, which masks the highest-saliency tiles (Eq. 3); and AICD, which masks the lowest-saliency tiles (Eq. 4). Both use $\rho = 0.5$ for clear visualisation and a Gaussian-blur fill. The same saliency map produces both masks.

clean. A lower ρ is used only for the visualisations in this paper, where a larger masked area is easier to read.



Figure 2: Four of the five fill strategies of Table 1 (the `global_mean` strategy is omitted for space) applied under a fixed mask, for ICD (top row) and AICD (bottom row) on the same image. Gaussian blur is the default soft fill; local mean and noise preserve different aspects of the original content; solid zero-fill is the hard limit and reduces to classical Cutout.



Figure 3: Effect of the tile size t on the ICD mask (Gaussian-blur fill, $\rho = 0.5$). Smaller tiles follow the saliency map more closely at the cost of a fragmented occlusion; larger tiles yield a coarser, more contiguous mask.

4 Discussion

We motivate ICD and AICD through two distinct mechanisms, and then note the assumptions and costs that the construction carries.

4.1 ICD as model-adaptive difficulty

ICD removes exactly the evidence the model currently finds most convincing. Consider a classifier that has settled on a single dominant cue, such as a dog’s face or a flower’s centre. Under ICD it repeatedly encounters that cue occluded and must predict from what remains. This is a form of difficulty that adapts to the model rather than to the data: the same image presents a harder problem to a network that relies narrowly on one region than to one that already integrates several. The mechanism is related to the motivation of curriculum learning [Bengio et al., 2009], with the ordering inverted. Where a curriculum presents easy examples first and hard examples later, ICD holds the data fixed and instead removes, at each step, whatever the model has found easy. The target it removes therefore moves, defined by the model’s own saliency. Because the saliency in Eq. (1) is recomputed from the current parameters, the occlusion follows the representation as it develops rather than being fixed in advance.

4.2 AICD as background suppression

AICD addresses the complementary failure mode. In many natural datasets the background is statistically informative: water co-occurs with certain animals, indoor scenes with certain objects [Beery et al., 2018]. A model can exploit this without ever attending to the object, and such reliance is invisible to validation accuracy when the validation set shares the same background statistics. AICD masks precisely the

low-saliency region, which is the background once the model has localised the object, and perturbs it with a fill. Repeatedly altering the background while preserving the object discourages the classifier from treating background as evidence, in the spirit of work that removes or randomises background to test and improve object-level reliance [Xiao et al., 2021].

4.3 Expected effect on saliency

Both methods act on the saliency map and may, in turn, reshape it. Suppressing the high-saliency region (ICD) should discourage the concentration of saliency on any single area, so we would expect a model trained with ICD to develop a more distributed saliency map; suppressing the background (AICD) should, if it succeeds, sharpen saliency onto the object. These expectations are stated here as hypotheses. Quantitative measures such as the fraction of saliency mass on the image border, the coverage above a threshold, and the Gini concentration of the map can make them testable, and we take up that measurement in the companion study rather than claiming any outcome here.

4.4 Assumptions and limitations

The construction rests on several assumptions that bound its applicability.

Saliency must be meaningful. ICD and AICD are only as good as the saliency map that drives them. At initialisation, before the network has learned useful features, the GradCAM map is close to uninformative, and masking by it approaches random masking. The methods are therefore best applied once the model has partially trained, or with saliency computed from a warmed-up checkpoint; applying them from the first step is unlikely to differ from Cutout.

Online cost. Unlike dataset-level augmentations that can be precomputed, the saliency in Eq. (1) depends on the current parameters and in principle must be recomputed as training proceeds. A gradient evaluation per image per update is expensive. Section 5 describes the caching strategy we use to amortise this, but the cache trades freshness for speed: a cached map reflects the model at the time of caching, not the current step.

Tile granularity. The tile size t is a fixed grid imposed on a continuous saliency map. A small t can mask fine object parts that happen to be salient; a large t can mask substantial background along with the object. The appropriate t depends on the object scale relative to the image, which the method does not adapt automatically.

Single-image, single-label scope. ICD and AICD operate on one image and leave the label unchanged, which keeps them simple and composable but also means they do not exploit the label-mixing that benefits methods such as CutMix [Yun et al., 2019] and SnapMix [Huang et al., 2021]. The construction as presented is specific to classification; extending the tile-scoring step to detection or segmentation, where saliency is defined per region or per pixel, is left to future work.

5 Implementation

ICD and AICD are implemented in the open-source BNNR library [Walo et al., 2026] (MIT licensed, available on PyPI as `bnnr`). The saliency back end is the `pytorch-grad-cam` package [Gildenblat and contributors, 2021], so the maps of Eq. (1) come from a standard, widely used GradCAM implementation rather than a bespoke one; by default the library enables that package’s eigen-smoothing option, as noted in Section 3.1. This section describes how the methods are applied during training.

5.1 Online application and the saliency cache

Because the saliency map depends on the current parameters, a naive implementation would run a GradCAM pass for every image at every step, roughly doubling the cost of training. BNNR amortises this with a disk-backed cache. Before a training run (or after a warm-up phase, as discussed in Section 4.4), the saliency map for each training image is computed once and stored, keyed by a hash of the image content:

```

from bnnr.xai_cache import XAICache
cache = XAICache("./xai_cache")
cache.precompute_cache(
    model=model,
    train_loader=train_loader, # yields (image, label, index)
    target_layers=target_layers, # e.g. [model.layer4[-1]]
    n_samples=len(train_dataset),
    method="gradcam",
)

```

During training the augmentations read tile masks from the cache instead of recomputing GradCAM, reducing the per-step overhead to a cache lookup and the masking arithmetic of Eqs. (2)–(6). The cache reflects the model at the time of precomputation; refreshing it periodically trades additional compute for saliency that tracks the evolving model more closely.

5.2 Use in a training loop

The augmentations are plain callables that take a batch and its labels. A classification step looks as follows:

```

from bnnr.icd import ICD, AICD
icd = ICD(model=model, target_layers=target_layers,
           cache=cache, threshold_percentile=70.0,
           tile_size=8, fill_strategy="gaussian_blur",
           probability=0.5)
aicd = AICD(model=model, target_layers=target_layers,
             cache=cache, threshold_percentile=70.0)

for images, labels, index in train_loader: # images uint8 NHWC
    images = icd.apply_batch_with_labels(
        images, labels, sample_indices=index)
    # ... convert to [0,1] BCHW, forward, backward, step ...

```

The label passed to the augmentation is the class whose saliency defines the mask; by default this is the ground-truth label, though the model’s prediction may be used instead. The `DataLoader` yields a sample index as a third element so the cache can retrieve the correct precomputed map. Images are expected in `[0, 1]` (or `uint8`, converted internally); ImageNet normalisation, if used, is applied after the augmentation so that the saliency back end and the fill operate on unnormalised pixels.

ICD and AICD are independent augmentations and need not be used together. Each can be inserted into any existing PyTorch loop, including framework wrappers such as Lightning, without further dependencies, and either can be composed with conventional augmentations (random crop, flip, colour jitter) applied before or after the masking step.

6 Conclusion

We have presented Intelligent Coarse Dropout (ICD) and Anti-ICD (AICD), two augmentation methods that turn a GradCAM saliency map into a tile-based occlusion mask. ICD masks the tiles the model finds most salient, removing the evidence it currently relies on; AICD masks the least salient tiles, perturbing the background while leaving the discriminative region intact. The two are governed by a single percentile threshold that selects them as opposite directions of the same construction and fixes their occlusion area to a common fraction. Masked tiles are replaced by one of several soft fills rather than a hard box, and the masking operates at a coarse tile granularity that tempers the noise of pixel-level saliency. The methods are implemented in the open-source BNNR library on top of `pytorch-grad-cam`, with a saliency cache that makes their online use practical.

The present paper introduces the methods and their rationale; it makes no empirical claim about their effect on accuracy. The natural next step is a controlled comparison of ICD and AICD against RandAugment, TrivialAugment, AutoAugment, and random masking, under matched compute and a held-out test split. That comparison, together with the saliency measurements proposed in Section 4.3, would test whether the methods reshape the model’s attention as their design intends. We report that

study separately. Further directions include adapting the tile size to object scale, refreshing the saliency cache on a schedule, substituting alternative class activation methods for GradCAM, and extending the tile-scoring construction beyond classification to detection and segmentation.

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